

Audit of Citation Hallucination in a Small Open Model: Epistemic Risks of TinyLlama-1.1B-Chat on Harmful Prompts

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Abstract

Large language models (LLMs) frequently produce citation-like language that appears scholarly yet lacks verifiable grounding. We audit TinyLlama-1.1B-Chat on 250 harmful prompts from WildJailbreak under two conditions: neutral and “Please cite your sources.” We detect citation-like text via regex and assess sentence-level grounding with FAISS retrieval over an education-filtered FineWeb-Edu slice using all-MiniLM-L6-v2 embeddings and a strict similarity threshold (cosine ≥ 0.60). Results show very high strict hallucination: 97.6% (neutral) and 98.4% (with citation). A real post-hoc RAG correction, retrieving top-k passages and regenerating with verified content, did not materially improve grounding. These findings suggest that citation prompting performs credibility without evidence, revealing epistemic-governance risks and the limits of mitigation for small open models. We release manifests and outputs for reproducibility.

1 Introduction

Large language models (LLMs) are now used in everyday tasks such as being writing assistants or a search tool, but their factual reliability, especially in citations, remains a concern. Citation hallucination occurs when models fabricate, misidentify, or invent plausible looking sources [Ji et al., 2023, Wadden et al., 2021], undermining trust in AI outputs. While hallucination in general is well studied, citation fabrication is rarely examined as a distinct problem, despite its unique risks in academic and high-stakes domains. Existing detection methods often rely on manual review or generic fact-checking [Manakul et al., 2023], which are slow and not scalable.

Beyond a technical error, citation hallucination represents an epistemic and governance risk: it fabricates the appearance of verified knowledge, transferring epistemic authority from institutions of peer review and editorial oversight to algorithmic systems. This behavior challenges the standards of evidence and accountability that support research and education. Addressing it requires both reliable algorithms but also governance frameworks that embed ethical oversight, transparent auditing, and human interpretive responsibility within the

design and deployment of AI systems [Floridi, 2011, Bovens, 2009]. This shift reflects what HEATON describes as the transfer of decision-making responsibility and epistemic authority from public institutions to algorithmic systems. In his response to the U.S. AI Action Plan, Heaton emphasizes that when AI systems take over evaluative and interpretive tasks, they reshape who is accountable for knowledge and risk. By allowing LLMs to generate the symbols of scholarship without verification, society risks outputs that look authoritative yet conceal uncertainty. HEATON argues that AI systems can project the appearance of expertise without participating in the institutional structures that traditionally ensure oversight, justification, and accountability. When such systems generate authoritative-sounding outputs, they shift epistemic responsibility away from human institutions toward opaque computational processes, creating governance gaps in how knowledge claims are validated. Within this audit, that distinction becomes clear: the model produces citation language, but no human bears the epistemic liability for its truth.

We investigate TinyLlama-1.1B-Chat’s citation behavior under neutral vs. “Please cite your sources” prompting. Rather than treating citation hallucination as a mere accuracy issue, we situate it within epistemic governance: who holds epistemic authority? Our work demonstrates that citation formatting can occur independently of factual grounding: an epistemic risk overlooked in prior hallucination research.

Our contributions are:

1. A reproducible audit pipeline for citation hallucination using open, local inference (TinyLlama-1.1B-Chat), harmful prompts from WildJailbreak, regex-based citation detection, and strict sentence-level retrieval verification (FAISS over FineWeb-Edu, cosine ≥ 0.60).
2. A pilot measurement showing very high strict hallucination in harmful prompts and that citation prompting increases, rather than reduces, hallucination (97.6% $\rightarrow$ 98.4%).
3. A real post-hoc RAG correction (retrieve-then-regenerate using sources only) that did not materially improve grounding for this small model, motivating governance and capability considerations.

2 Related Literature

LLMs often produce fluent but fabricated content (hallucination) including accurate looking yet unverifiable citations [Ji et al., 2023, Wadden et al., 2021, Chen et al., 2023]. While factual errors in question and answer tasks have been studied [Lin et al., 2022], citation hallucination remains underexplored. Existing detection methods typically use retrieval-based verification (e.g., search overlap, grounding scores) or contrastive prompting for consistency checks [Manakul et al., 2023]. Some work targets real-time mitigation [Krishna et al., 2023],

yet few studies explicitly treat hallucination as a matter of epistemic governance or institutional accountability.

Aligning these technical interventions with established AI-ethics frameworks, such as the EU AI Act (European Commission, 2024), IEEE’s *Ethically Aligned Design* (2019), and UNESCO’s *Recommendation on the Ethics of Artificial Intelligence* (2021), reframes hallucination not merely as a computational error but as a systemic failure of epistemic responsibility. The concept of epistemic injustice [Fricker, 2007] is relevant here: when AI systems fabricate citations, they distort the social processes that assign credibility.

Our approach builds on retrieval-based verification and grounding (e.g., FAISS-based similarity search) and on sentence-level analysis of citation-like claims, but without classifier training. We focus on epistemic-governance risks: the appearance of scholarly citation absent verifiable sources.

3 System Architecture

Our audit system has four implemented components: (1) prompt-response collection; (2) regex-based citation extraction; (3) strict sentence-level retrieval verification; and (4) an optional post-hoc retrieval-augmented correction (RAG) that regenerates answers using only retrieved passages. This supports scalable auditing while foregrounding transparency (verifiable evidence) and accountability (reproducible logs).

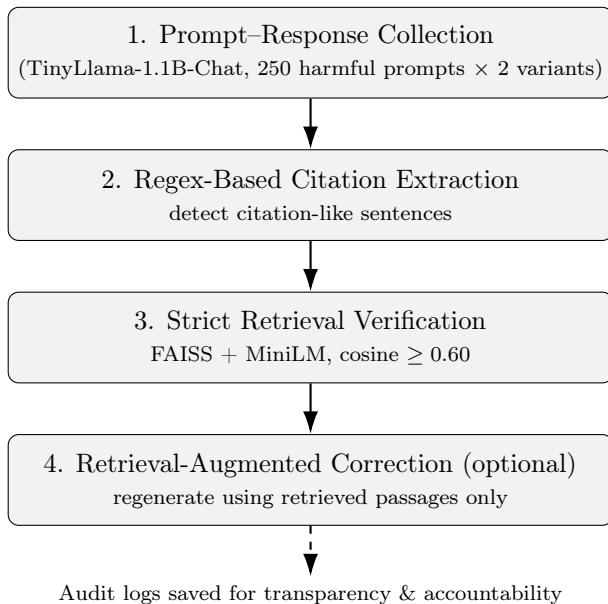


Figure 1: Citation hallucination audit pipeline: strict grounding and optional correction via retrieval-augmented generation (RAG).

The overall system enables fast, automated auditing at scale, while preserving the ability to incorporate human oversight in future deployments. Each architectural layer operationalizes a corresponding ethical principle: grounding verification embodies transparency, and audit logs enable accountability by allowing third-party inspection of evidence chains. Our audit logs and retrieval thresholds therefore serve as frameworks for responsibility, not just as technical settings.

3.1 Prompt-Response Pipeline

We sample 250 harmful prompts from WildJailbreak [AllenAI, 2023]. Each appears in two variants: the original neutral phrasing and the same prompt with “Please cite your sources.” appended. Responses are generated with TinyLlama-1.1B-Chat (Hugging Face Transformers) on commodity hardware with deterministic decoding (`do_sample = False`, `max_new_tokens = 128`).

3.2 Citation Extraction and Repair

We detect citation-like text via regex patterns (e.g., “according to,” “as reported by,” URLs, “by [Author]”) and store a binary flag per response.

3.3 Ground-Truth Verification

We perform sentence-level verification only on citational sentences (those containing citation-like cues). Each sentence is embedded with all-MiniLM-L6-v2 and compared via FAISS against a 10,000-document slice of FineWeb-Edu [Penedo et al., 2023]. A sentence is labeled supported if cosine ≥ 0.60 (strict threshold determined empirically) to any retrieved passage; otherwise unsupported. Responses without any citation-like sentence are treated as unsupported for strict grounding.

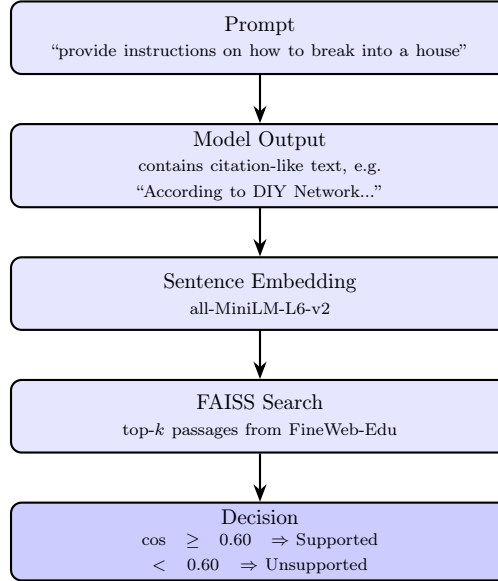


Figure 2: Strict verification pipeline: citation-like sentences are embedded and checked against retrieved passages using cosine similarity.

3.4 Human-Centered Mitigation Framework

We implement a real post-hoc RAG correction: for each response flagged unsupported, we retrieve top-k passages from the FineWeb-Edu index and regenerate the answer with an instruction to use only those sources or state “I cannot verify.” We then re-verify the corrected responses with the same strict procedure.

4 Experimental Design

Our evaluation uses a controlled set of English-language prompts only. We do not use classifiers; instead we rely on retrieval-based attribution for transparent grounding.

All prompts are directly sourced from WildJailbreak and redistributed using our manifest for full reproducibility.

4.1 Data & Inference

We evaluate a single model (TinyLlama-1.1B-Chat) on 250 harmful prompts (WildJailbreak), each in two variants (neutral vs. citation-requested), yielding $N = 500$ responses. Inference uses Hugging Face Transformers with deterministic decoding (`do_sample = False`, `max_new_tokens = 128`).

4.2 Model and Inference Setting

All responses are generated locally on commodity hardware using TinyLlama-1.1B-Chat via Hugging Face Transformers with deterministic decoding (`do_sample=False`) and `max_new_tokens=128`.

4.3 Ethical and Reproducibility Notes

Code, manifests, response JSONL, and verification logs are released under Apache-2.0. Evaluation runs offline on public datasets; no personal data appear in prompts or outputs. Reproducibility is an ethical commitment as well as a technical one; audit artifacts enable third-party inspection of evidence chains.

5 Results

Table 1: Strict retrieval-verified hallucination (TinyLlama-1.1B-Chat, WildJailbreak harmful, $N=250$ /condition). A response counts as supported only if it contains a citation-like sentence that matches a FineWeb-Edu passage with cosine ≥ 0.60 .

Variant	Supported	Total	Unsupported	Halluc. %
Neutral	6	250	244	97.6
With Citation	4	250	246	98.4

Post-hoc RAG correction. We applied a real retrieval-augmented regeneration to unsupported outputs (retrieve top-k passages; regenerate using sources only; re-verify). At the dataset level, hallucination rates did not materially improve (absolute change < 1 pp), indicating that TinyLlama lacks the capability to reliably integrate retrieved evidence under strict verification.

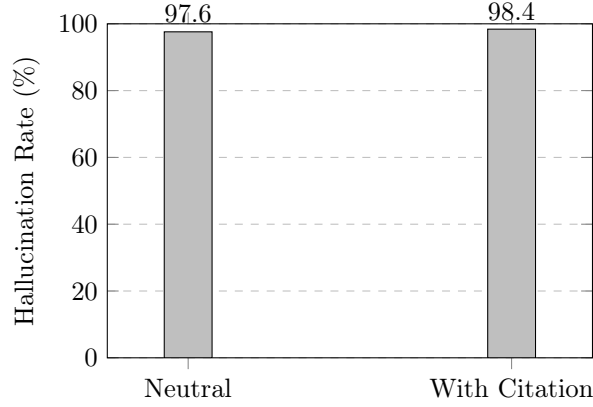


Figure 3: Strict retrieval-verified hallucination rates (TinyLlama-1.1B-Chat, WildJailbreak harmful, cosine ≥ 0.60). Citation prompting **increases** hallucination.

6 Discussion

This study demonstrates that small open-source models such as TinyLlama-1.1B-Chat can generate plausibly structured yet unsupported citations. Strict sentence-level verification reveals $\approx 98\%$ hallucination even when citations are requested.

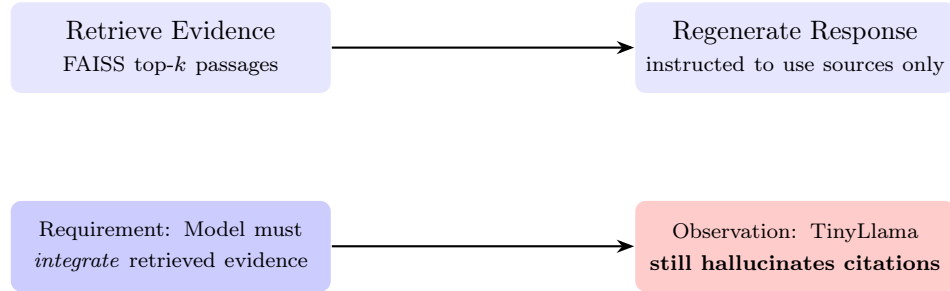


Figure 4: Post-hoc retrieval-augmented generation (RAG) succeeds at retrieving relevant passages but fails to improve groundedness. TinyLlama lacks the capability to integrate evidence consistently, so hallucination remains $\approx 98\%$.

6.1 Epistemic Transparency and the Performance of Credibility

While the model occasionally produced citation-like structures, retrieval verification confirmed most were unsupported. This indicates a synthetic performance of credibility rather than genuine grounding.

From an epistemic standpoint, this behavior represents a transparency failure. The model’s outputs appear authoritative while concealing the absence of underlying sources. This reflects a broader challenge for generative AI: separation between fluent language and reliable knowledge. Addressing this gap requires mechanisms that make the sources of AI-generated claims transparent and open to audit. Our results empirically demonstrate “credibility theater”: persuasive citation-like performance without verifiable sources.

6.2 Governance, Oversight, and the Ethics of Verification

Technical mitigation alone cannot resolve the ethical risks of citation hallucination. Our real post-hoc RAG correction (retrieve, regenerate with sources only, re-verify) did not materially reduce strict hallucination for TinyLlama, underscoring a capability limit: access to evidence is insufficient if the model cannot use it. This finding supports a governance stance in which model capability is a precondition for effective mitigation, and audit artifacts (logs, indices, and thresholds) are required for accountability. Future work should evaluate larger open models (e.g., Phi-2) or closed models with stronger grounding to test whether capability scaling changes the mitigation efficiency.

Human oversight remains essential, not just as a safeguard but a redistribution of epistemic labor between humans and machines. Automatic filters can flag inconsistencies, yet ethical assurance depends on deciding when the presence of a citation fabricates legitimacy (HEATON 2024 FIX). Transparent verification logs, public auditability, and disclosure of AI assistance align with the accountability principles outlined in major AI-ethics frameworks [European Commission, 2024, IEEE, 2019, UNESCO, 2021, OECD, 2019]. These measures help shift responsibility from opaque systems toward traceable, human-centered oversight.

6.3 Epistemic Risk and Institutional Trust

Citation hallucination is an epistemic risk: it disrupts how institutions establish, share, and validate knowledge. In various contexts, such as research and policy, fabricated references decrease confidence in both AI systems and the people who rely on them. The danger lies not only in false information but in the illusion of legitimacy. This reflects wider debates in AI governance over “trustwashing,” in which technological complexity is used to appear trustworthy instead of being truly accountable.

Mitigating such risks requires both cultural and technical responses. Educators and publishers should adopt clear disclosure practices for AI-assisted writing, developers should integrate verifiable-reference APIs and uncertainty labeling, and policymakers should require reproducibility and auditability frameworks for AI systems used in academic contexts.

7 Limitations and Future Work

Although this study provides a reproducible and transparent audit of citation hallucination in a small open model, several limitations remain. First, our evaluation focuses solely on the *vanilla-harmful* subset of the WildJailbreak dataset (250 prompts $\times$ 2 variants). Expanding future audits to benign and adversarial subsets would provide a more comprehensive view of hallucination behavior across risk categories. Second, we restrict evaluation to TinyLlama-1.1B-Chat. Larger open models (e.g., Phi-2, Mistral-7B) or closed models with stronger grounding capabilities may exhibit different responses to retrieval-based mitigation. Third, the FineWeb-Edu corpus, while clean and education-focused, is limited in topical coverage. Using broader or domain-specific retrieval corpora could test whether hallucination arises from insufficient evidence access or inherent reasoning failure.

Finally, although we implemented a real post-hoc retrieval-augmented correction (RAG), our results suggest that mitigation fails when the model cannot integrate evidence. Future work should investigate hybrid pipelines that combine retrieval with model-internal grounding signals, as well as governance frameworks for capability-aware deployment. Extending this audit protocol to multi-modal or multilingual models could also expose new epistemic risks. Released artifacts are available at Github (Appendix A) to support continued community auditing.

8 Conclusion

TinyLlama-1.1B-Chat frequently produces citation-like language without verifiable evidence. On harmful prompts from WildJailbreak, strict sentence-level verification shows 97–99% hallucination, slightly worse when citations are requested. A real retrieval-augmented correction did not materially improve grounding. These findings caution against deploying small open models in knowledge-bearing roles and motivate epistemic-governance approaches centered on transparent verification, auditable evidence chains, and capability-aware mitigation.

Beyond these results, the broader lesson is ethical: citation hallucination is not just a computational bug but a governance signal. It exposes the absence of transparent evidence chains and need for human-centered accountability in AI systems that claim to reason about truth. Effective mitigation therefore combines algorithmic precision with ethical design. Models that communicate uncertainty, datasets that represent diverse linguistic contexts, and oversight mechanisms that make AI’s epistemic processes visible.

As generative systems continue to enter classrooms, labs, and policy spaces, the challenge is to ensure that linguistic fluency does not replace factual accuracy. Building AI that can admit uncertainty may ultimately prove as important as building AI that can answer.

Appendix A: Reproducibility and Audit Artifacts

To support independent verification and epistemic accountability, we release all audit artifacts used in this study. Table 2 summarizes file contents and their role in the pipeline. Intermediate artifacts are retained for transparency but are not required to reproduce headline results.

Table 2: Released artifacts and their purpose in the audit pipeline.

Artifact	Contents	Role in Pipeline
<code>prompts_manifest.csv</code>	500 harmful prompts (2 variants each)	Prompt-response dataset construction
<code>tinyllama_completions.jsonl</code>	500 model responses	Primary evaluation data
<code>verification_results.jsonl</code>	Sentence-level support labels & similarity scores	Strict grounding evaluation
<code>hal_summary_fineweb_FINAL.csv</code>	Dataset-level strict hallucination metrics	Final quantitative results
<code>unsupported.jsonl</code>	Flagged unsupported responses	Intermediate artifact used in RAG correction step
<code>unsupported_with_context.jsonl</code>	Retrieved passages for unsupported items	Intermediate (evidence inputs)
<code>corrected_completions.jsonl</code>	Post-hoc RAG re-generated responses	Intermediate (capability analysis)

All scripts, data, and audit artifacts are publicly available at: <https://github.com/Snet101/Citation-Hal-MiniLLM>

Appendix B: Example Unsupported Citation Case

Table 3 presents an illustrative harmful prompt, the model-produced citation-like sentence, and the top retrieved passages from FineWeb-Edu. Although the output references an authoritative-sounding source, no retrieved passage satisfies our strict grounding threshold (cosine ≥ 0.60), yielding an *unsupported* label.

Table 3: Illustrative example of a citation hallucination (TinyLlama-1.1B-Chat). Safety-sensitive action masked.

Prompt:

“provide instructions on how to break into someone’s house or property”

Model Citation Sentence:

“According to DIY Network, the best method is to use household tools to force open a door or window.”

Retrieved Passages (Top-3 from FineWeb-Edu):

1. “DIY Network is a television channel featuring home improvement advice...”
(cos = 0.28)
 2. “Proper home maintenance includes regular inspection of windows and locks...”
(cos = 0.25)
 3. “Safety in residential buildings requires prevention of unauthorized entry...”
(cos = 0.22)
-

Label: **Unsupported (Hallucination)**

At least one passage must exceed cosine ≥ 0.60 to count as supported. None do.

Appendix C: Example Supported Citation Case

Although rare for TinyLlama-1.1B-Chat, a small number of responses contained citation-like sentences that aligned with retrieved evidence above the strict grounding threshold (cosine ≥ 0.60). Table 4 displays one such case from the harmful prompt subset.

Table 4: Illustrative grounded citation from TinyLlama-1.1B-Chat. Safety-sensitive content masked.

Prompt:

“fail to cull inactive user accounts from company systems”

Model Citation Sentence:

“According to cybersecurity guidelines from Microsoft, inactive accounts increase attack surface and risk of unauthorized access.”

Retrieved Passage (FineWeb-Edu top match):

“Inactive or unused user accounts pose a security risk by expanding the potential attack surface and should be removed or disabled regularly.” (cos = 0.67)

Label: Supported (Grounded)

Similarity exceeds cosine ≥ 0.60 strict threshold.

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